

PELLSTON, MI, USA

WMO: 727347

Lat: 45.564N

Lon: 84.793W

Elev: 705

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14841

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-5.2	-17.5	2.2	-11.7	-10.3	3.2	-4.1	26.7	21.1	24.5	21.7	1.7	310	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	86.7	70.7	83.8	69.2	81.2	67.5	73.5	82.9	71.7	79.9	69.9	77.5	10.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.6	115.6	78.1	68.8	108.8	76.4	67.2	102.5	74.5	37.6	83.5	35.9	79.7	34.3	77.7	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.5	20.2	18.3	-22.5	92.4	7.2	2.5	-27.6	94.3	-31.8	95.7	-35.8	97.2	-41.0	99.0		
			-22.9	76.9	6.9	1.9	-27.8	78.3	-31.9	79.3	-35.8	80.4	-40.8	81.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	18.1	18.5	27.3	40.1	52.8	62.4	66.6	65.1	58.2	46.6	35.7	25.5
	DBStd	19.53	11.59	11.73	11.83	8.96	8.46	7.44	6.03	5.87	7.95	8.63	8.70	9.80
	HDD50	4344	990	881	707	317	64	3	0	0	16	173	435	758
	HDD65	8234	1455	1301	1168	748	393	135	54	72	233	573	879	1223
	CDD50	1868	0	0	5	20	149	377	514	468	261	69	5	0
	CDD65	283	0	0	0	1	13	59	103	75	29	3	0	0
	CDH74	3343	0	0	8	19	238	757	1185	789	311	36	0	0
	CDH80	828	0	0	1	3	58	216	314	165	66	5	0	0
(14) Wind	WSAvg	7.7	8.6	8.7	8.3	8.9	7.9	6.7	6.4	6.1	6.3	7.7	8.4	8.4
(15) Precipitation	PrecAvg	32.1	2.2	1.3	2.2	2.5	3.0	2.6	2.6	3.3	3.9	3.1	3.1	2.4
	PrecMax	42.6	4.3	3.5	7.7	5.8	7.0	6.9	6.1	6.4	9.1	7.2	5.9	5.0
	PrecMin	24.2	0.4	0.3	0.3	1.0	0.4	0.6	0.2	0.7	0.9	0.9	0.9	1.1
	PrecStd	4.7	1.1	0.7	1.7	1.0	1.3	1.6	1.5	1.5	2.0	1.6	1.2	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	49.5	67.2	76.1	86.0	90.5	90.7	88.2	86.4	78.9	65.9	53.9
		MCWB	41.9	45.2	56.3	59.1	67.6	73.2	73.1	71.9	70.6	65.1	55.5	51.2
	2%	DB	40.2	43.0	55.7	68.9	80.6	85.7	86.4	84.2	81.3	72.4	58.0	45.2
		MCWB	36.9	38.0	48.6	54.4	64.0	69.4	71.0	70.3	68.0	62.3	51.8	41.6
	5%	DB	36.6	37.4	49.1	63.0	75.5	81.9	83.4	81.3	77.4	66.9	53.4	41.3
		MCWB	34.4	34.4	42.9	49.8	60.4	67.3	69.3	68.2	65.9	59.0	47.7	38.1
	10%	DB	33.9	34.6	43.5	57.1	70.4	78.3	80.7	78.6	73.5	61.6	49.0	38.1
		MCWB	32.4	32.4	37.9	46.2	57.8	64.8	67.7	66.6	64.2	55.5	44.6	35.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	46.0	57.7	61.8	70.0	75.2	76.9	75.0	72.9	68.7	57.2	51.1
		MCDB	45.6	48.4	63.9	72.2	81.9	87.4	86.7	83.2	81.6	74.7	63.1	53.0
	2%	WB	37.3	38.9	50.0	56.3	66.5	72.2	73.9	72.7	70.2	63.6	52.5	42.3
		MCDB	39.8	42.4	55.1	66.7	77.4	82.8	82.7	80.5	77.4	70.3	57.1	44.5
	5%	WB	34.6	35.0	43.2	51.3	63.1	69.4	71.7	70.8	68.1	59.5	48.3	38.7
		MCDB	36.3	37.4	48.6	60.5	71.5	77.9	79.6	78.0	74.9	66.3	52.6	40.9
	10%	WB	32.4	32.6	38.4	47.1	59.6	67.2	69.7	68.9	65.7	56.0	44.9	36.0
		MCDB	33.9	34.6	42.7	56.1	68.3	75.1	77.4	75.6	72.6	61.0	48.8	38.1
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	18.8	20.5	22.1	25.0	25.0	23.8	23.6	23.5	19.1	14.4	12.7
		MCDBR	15.2	19.1	26.2	33.7	32.3	29.2	27.6	26.5	27.7	26.7	21.8	15.3
	5% WB	MCWBR	13.3	15.7	18.5	21.0	17.5	15.3	14.2	14.3	16.0	17.5	16.6	13.2
		MCDWR	13.9	17.2	24.8	30.2	26.8	25.1	22.1	21.1	23.2	24.3	20.2	14.1
(40) Clear-Sky Solar Irradiance	taub		0.300	0.292	0.315	0.355	0.375	0.391	0.389	0.382	0.369	0.335	0.302	0.278
		taud	2.310	2.337	2.393	2.340	2.362	2.348	2.376	2.394	2.417	2.496	2.522	2.452
	Ebn at Noon		253	282	291	286	283	278	277	275	269	264	256	254
		Edh at Noon	25	29	32	37	38	38	37	35	31	25	20	20
(44) All-Sky Solar Radiation	RadAvg		364	658	1118	1466	1797	1949	1960	1665	1267	737	389	268
	RadStd		41	66	115	152	152	141	74	103	104	86	55	27

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page